

→ Apply Cross Entropy loss?

$$X \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{#} 4$$

$$H_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{#} 4$$

$$H_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{#} 4$$

↓

$$H \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{#} 4 = 16$$

1. Transformer:

"The | cat | sat | on | a | wall"
 $\begin{matrix} [x_1] & x_2 & \dots & x_T \end{matrix}$
 $\mathbb{D} \times 1$

hidden size : $\mathbb{D}$

- size of the token vector
- width of the model
- matrix view

$$X \in \mathbb{R}^{T \times \mathbb{D}}$$

$X =$ $\begin{matrix} \text{The} & x_1 \\ \text{cat} & \vdots \\ \text{wall} & x_T \end{matrix} \begin{bmatrix} \dots & \dots \\ \vdots & \vdots \\ \dots & \dots \end{bmatrix}$ $\leftarrow$ each row, one token
 $T \times \mathbb{D}$

Sequence length : T

- Number of tokens
- number of rows in input matrix

of heads : N

- How many independent attention subspaces
- Split $\mathbb{D}$ into N parts : $\mathbb{D} = N \times H$

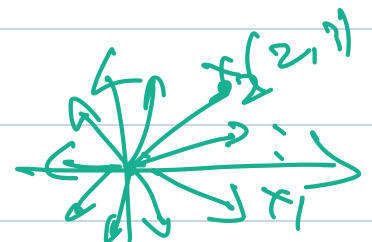
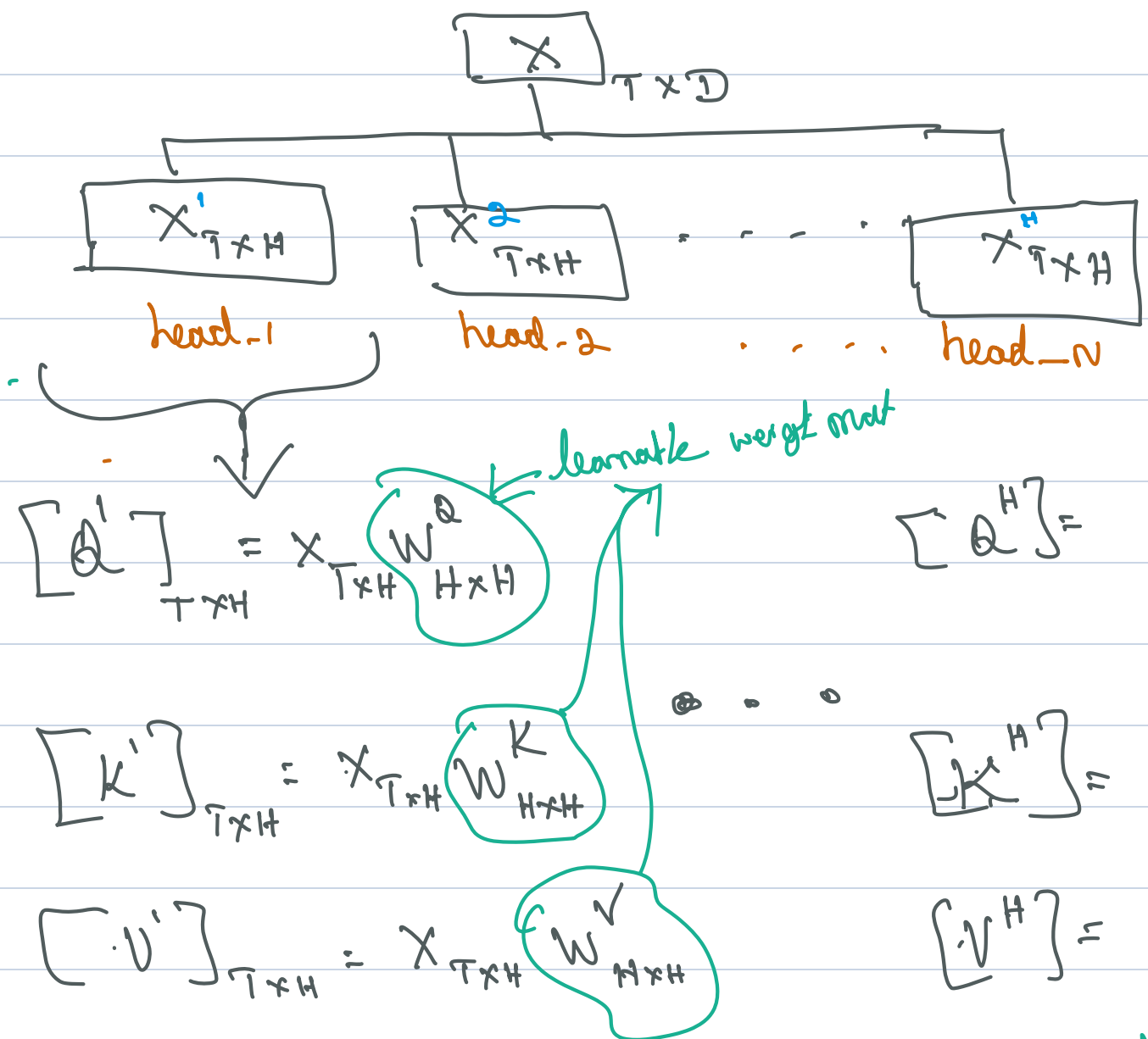
Head dimension : H [size of the W^Q , W^K & W^V matrices]

- Size of each head's subvector
- Each head attends in its own H -dim space

$X =$ $\begin{matrix} x_1 \\ \vdots \\ x_T \end{matrix} \begin{bmatrix} \dots & \dots \\ \vdots & \vdots \\ \dots & \dots \end{bmatrix} \begin{bmatrix} \dots & \dots \\ \vdots & \vdots \\ \dots & \dots \end{bmatrix} \dots \begin{bmatrix} \dots & \dots \\ \vdots & \vdots \\ \dots & \dots \end{bmatrix}$
 $T \times H$ $T \times H$ $T \times H$ $T \times \mathbb{D}$
 $T \times H$

> First head See all $\{T_i\}$ tokens but only the first "H" features in each token!

2. Q, K & V matrices:



3. Attention score

$$\text{Score}^1 = [Q' K'^T] = \begin{bmatrix} \quad \end{bmatrix}_{T \times T} \begin{bmatrix} \quad \end{bmatrix}_{T \times H} \begin{bmatrix} \quad \end{bmatrix}_{H \times T}$$

$$\begin{bmatrix} \text{attn}^1 \end{bmatrix}_{T \times H} = \begin{bmatrix} \text{softmax} \left(\frac{Q' K'^T}{\sqrt{D}} \right) \end{bmatrix}_{T \times T} \begin{bmatrix} V' \end{bmatrix}_{T \times H}$$

4. Merge all heads:

$$\begin{bmatrix} \text{attn} \end{bmatrix}_{T \times D} = \begin{bmatrix} \text{attn}^1 & \text{attn}^2 & \dots & \text{attn}^N \end{bmatrix}_{T \times D}$$

$\Downarrow$
 $[D = NH]$

5. MLP Layer

5.1 First linear (expansion)

typically the size of the expansion weight matrix will be $4 \times D (= F)$

in case of
modern transfor-
mers
2-expansions
used

$$\Rightarrow [X^{exp}]_{T \times F} = [X]_{T \times D} [W_{in}]_{D \times F}$$

learnable param
matrix

$$[X^{exp}]_{T \times F} = [X]_{T \times D} [W_{in}]_{D \times F}$$

$\Rightarrow D$ features has been increased by 4-times
ie to $F (4 \times D)$.

ie $D \rightarrow F$

5.2 Apply GELU to $[X^{exp}]_{T \times F}$

Apply SiGELU ($[X^{exp}]_{T \times F}$) $\leftarrow$ element wise

5.3 Second linear (Projection back)

$$[W]_{T \times D} = [X^{exp}]_{T \times F} [W_{out}]_{F \times D}$$

learnable matrix

6. Residual Connection

- Add the Residual to original Attention

$$\begin{bmatrix} X^{\text{new}} \end{bmatrix}_{T \times D} = \begin{bmatrix} \text{attn} X \end{bmatrix}_{T \times D} + \begin{bmatrix} \Delta W \end{bmatrix}_{T \times D}$$

LayerNorm: normalize across feature dim(D)

Second Attention layer.

Final transformer block.

$$H \in \mathbb{R}^{T \times D}$$

7. Linear Projection to Vocabulary

— To predict the next token, each hidden vector is projected into Vocabulary space. using the weight matrix

$$W^0 \in \mathbb{R}^{D \times N}$$

where N is the size of the Vocabulary.
for eg ($\approx 52k$)

— Compute logits: $\frac{\partial \mathcal{L}}{\partial W^0_{i,j}}$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\leftarrow$ split

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\leftarrow T \Rightarrow D$

$$[Z] = H_{T \times D} W^0_{D \times V} + b$$

$T \times V = 50k$

$$\begin{bmatrix} x_1 \dots x_{150k} \\ x_2 \dots x_{250k} \\ x_3 \dots x_{350k} \end{bmatrix}$$

$3 \times 50k$

8. Apply Soft-max Row-wise on Z

$$[P]_{T \times V} = \begin{bmatrix} \text{Softmax}(x_{1,1} \dots x_{1,V}) \\ \vdots \\ \text{Softmax}(x_{T,1} \dots x_{T,V}) \end{bmatrix}$$

$\Rightarrow e^{(0,1)}$

Eg:

$$[P] = \begin{bmatrix} x_1 & 0.1 & 0.05 & 0.7 & 0.01 & \dots \\ x_2 & \vdots & & & & \\ \vdots & & & & & \end{bmatrix}$$

Suppose the actual "Next Token" is y_t
 then the loss $L_1 = -\log(0.7)$

$$\Rightarrow \text{Sequence loss } L = \frac{1}{T} \sum_{t=1}^T -\log(P[t, y_t])$$

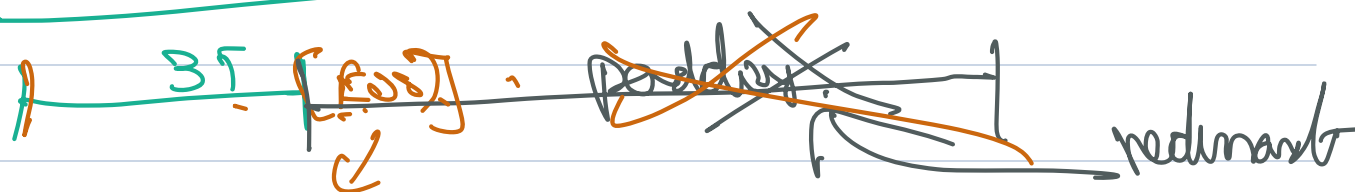
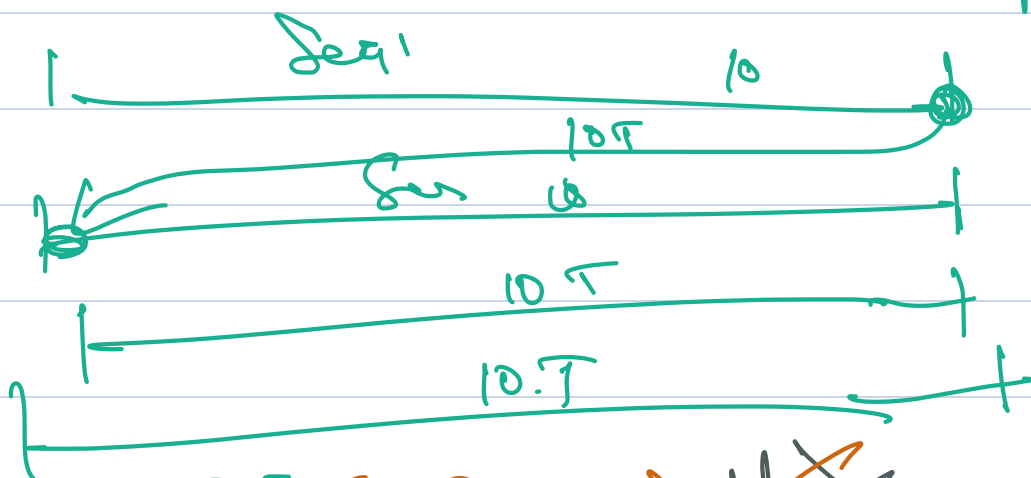
9. Input Vs Target

t	Input tokens	Target (y_t - next token)
1	The	cat
2	cat	sat
3	sat	on
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
T	mat	

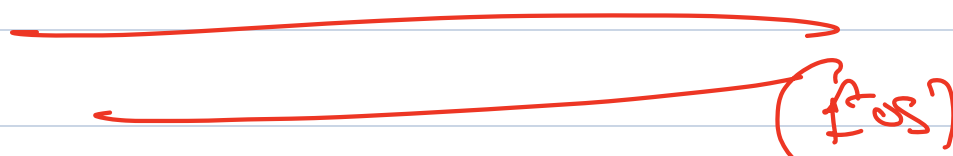
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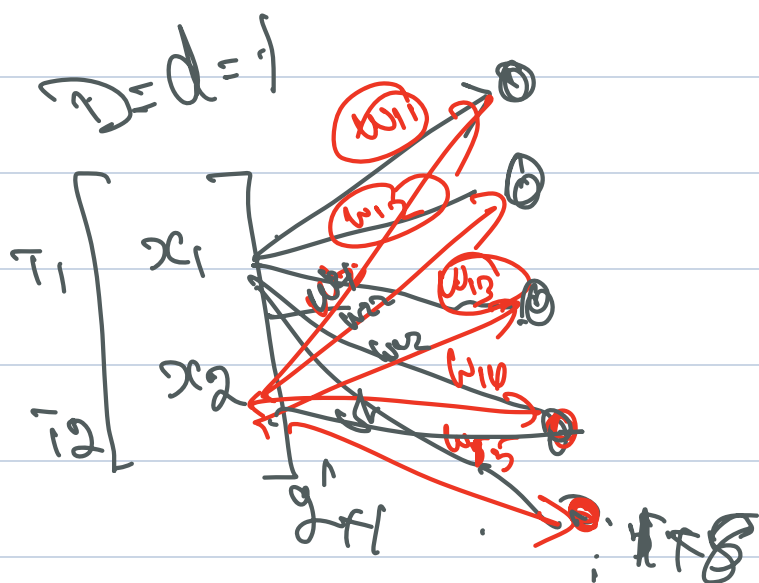
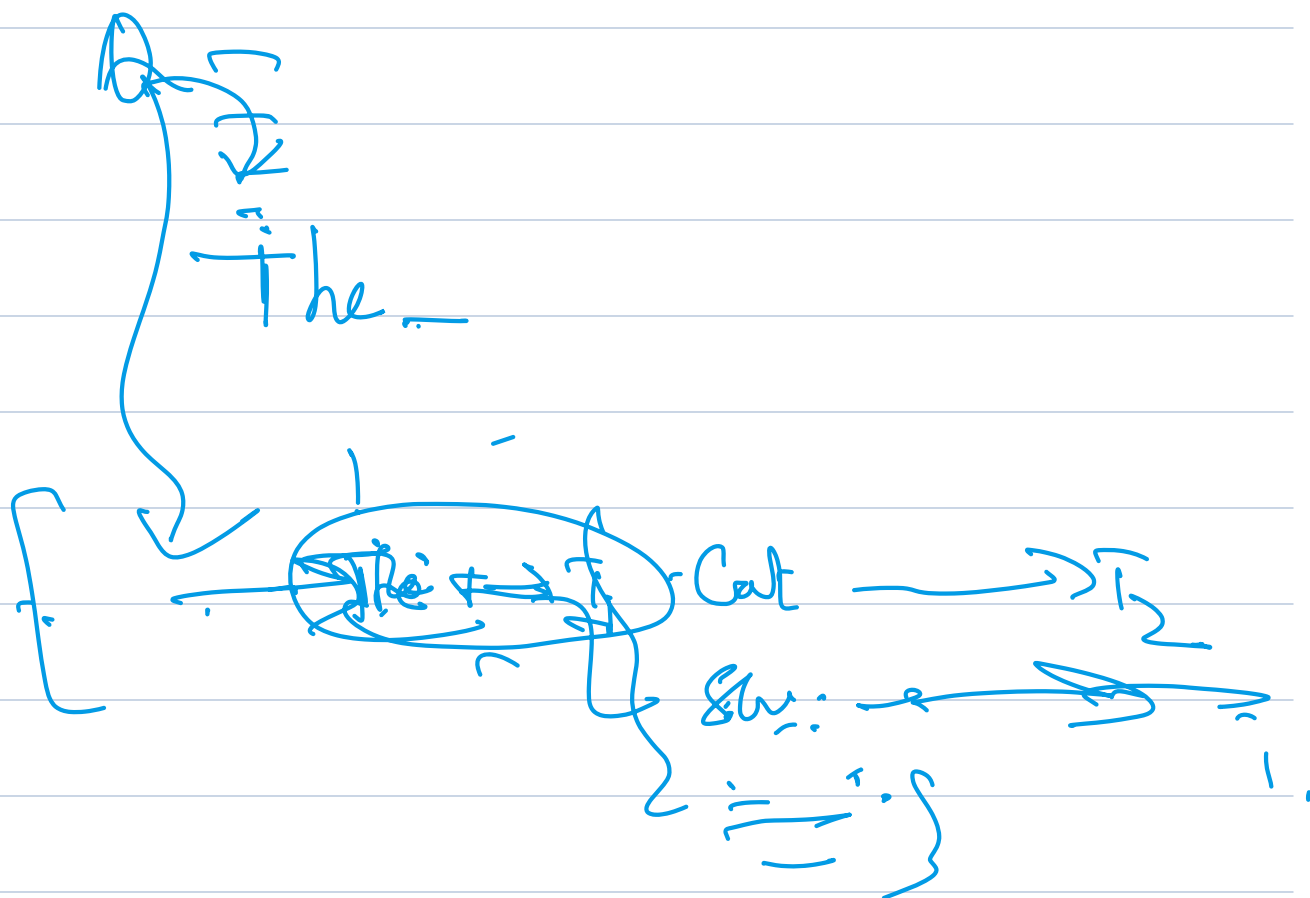
↑
EOS

Doc1



Doc2





Equation for the output T_2 :

$$T_2 = x_1 w_{11} + x_1 w_{12} + \dots + x_2 w_{21} + x_2 w_{22} + \dots + g_0 w_{g1} + g_0 w_{g2}$$

Equation for the output T_2 using matrix notation:

$$T_2 = \begin{bmatrix} x_1 & x_2 & g_0 \end{bmatrix} \begin{bmatrix} w_{11} & w_{12} & w_{g1} \\ w_{21} & w_{22} & w_{g2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{matrix}
 \downarrow \\
 \left[\begin{matrix} \vdots \\ \vdots \\ \vdots \end{matrix} \right]_{8 \times 1}
 \end{matrix}
 \quad
 \begin{matrix}
 w. \\
 \left[\begin{matrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ \vdots & \vdots \\ w_{81} & w_{82} \end{matrix} \right]
 \end{matrix}
 \quad
 \begin{matrix}
 \left[\begin{matrix} x_1 \\ x_2 \end{matrix} \right]
 \end{matrix}$$

$$\left[\begin{matrix} \vdots \\ \vdots \\ \vdots \end{matrix} \right]_{8 \times 1} = x_1 \left[\begin{matrix} w_{11} \\ \vdots \\ w_{81} \end{matrix} \right] + x_2 \left[\begin{matrix} w_{12} \\ \vdots \\ w_{82} \end{matrix} \right]$$